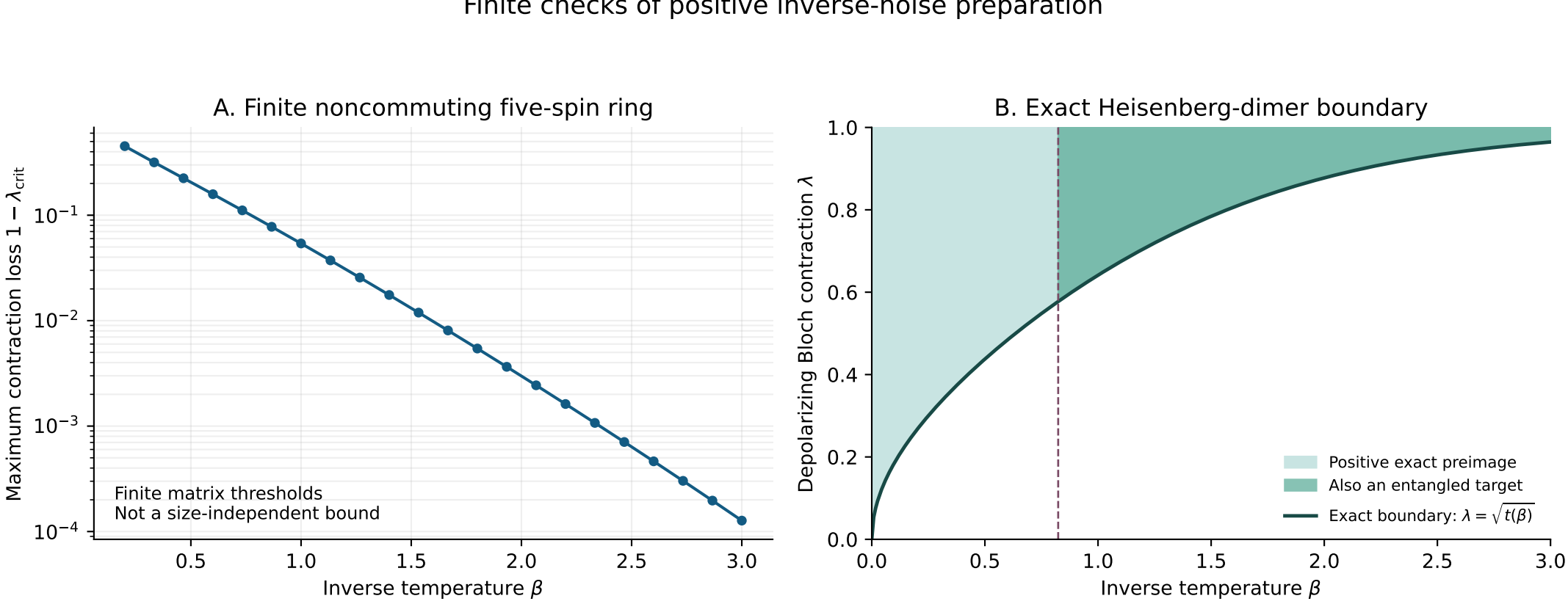
Finite checks of positive inverse-noise preparation
A. Finite noncommuting five-spin ring
Maximum contraction loss $1 - \lambda_{\mathrm{crit}}$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Finite matrix thresholds
Not a size-independent bound
0.5
1.0
1.5
2.0
2.5
3.0
Inverse temperature β
B. Exact Heisenberg-dimer boundary
Depolarizing Bloch contraction λ
1.0
0.8
0.6
0.4
0.2
0.0
Positive exact preimage
Also an entangled target
Exact boundary: $\lambda = \sqrt{t(\beta)}$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Inverse temperature β
Algebra and small matrices only; no source-sampler, apparatus, lifetime, or resource-advantage simulation.